

GPH-GU 2353 · Sample Warm-Up Quiz 1

Practice for Quiz 1 · Weeks 1–2 Topics · 15 Minutes · Multiple Choice

How to use this

This is a practice quiz. It is **not** the quiz you will sit, and none of these questions appear on it. It matches the real quiz in every other way: ten multiple-choice questions, one point each, fifteen minutes, and the same topics in the same proportions. If you can do this comfortably without help, you are ready.

Warm-Up Quiz 1 is on **Tuesday, September 22**, at the very start of class. It is **closed book, closed notes, and closed AI**, with no laptops, phones, or watches. It is on paper, so there is nothing to install and nothing to submit.

Practice it the way you will sit it. Put the slides away, close your browser, and do not run any R. Four of the ten questions come from Week 1 and six from Week 2, in no particular order. The answer key at the bottom names the topic each question was testing, so do all ten before you look.

Q1

A city health department fits a regression of hospital admissions on daily temperature, and uses it to anticipate how many beds will be needed during next week's heat wave. Which job is the model doing?

- A. Description
- B. Prediction
- C. Explanation
- D. None: regression cannot be used for planning

Q2

A magazine notes that athletes who appear on its cover usually perform worse the following season, and calls it a jinx. What is the statistical explanation?

- A. The added attention genuinely harms performance
- B. Athletes on covers are older than average
- C. Performance measures become less accurate over time
- D. Cover athletes are selected for an exceptional season, which is partly chance, so the next season is typically closer to their true average

Q3

Which of the following best defines a confounder in a study of x and y ?

- A. A variable measured with error
- B. An observation far from the fitted line
- C. A third variable that influences both x and y
- D. The part of y the model fails to capture

Q4

A regression of BMI on sleep hours adjusts for age, income, and physical activity. The adjusted association remains. Which level of the honest hierarchy have you reached?

- A. Association only
- B. Adjusted association, which supports causation only if the right confounders were measured
- C. Causation, since confounders were adjusted for
- D. Causation, since the sample is nationally representative

Q5

Why do we not choose the line that minimizes $\sum_i e_i$, the plain sum of the residuals?

- A. It has no closed-form solution
- B. It would give too much weight to large misses
- C. It is undefined when n is large
- D. Positive and negative residuals cancel, so a very poor line can achieve a sum of zero

Q6

In a sample, the correlation between sleep hours and BMI is $r = -0.30$, with $s_x = 1.5$ hours and $s_y = 6.0$ kg/m². Using $\hat{\beta}_1 = r \cdot s_y / s_x$, the slope is closest to:

- A. -0.075 kg/m² per hour
- B. -0.45 kg/m² per hour
- C. -1.20 kg/m² per hour
- D. -1.80 kg/m² per hour

Q7

A model of BMI on sleep hours gives a slope of -0.32 . Which sentence interprets it correctly?

- A. Adults who sleep one hour more have, on average, 0.32 kg/m² lower BMI
- B. Sleeping one more hour reduces BMI by 0.32 kg/m²
- C. Sleep explains 32% of the variation in BMI
- D. BMI falls by 32% for each additional hour of sleep

Q8

Adults in a sample report between 0 and 12 hours of vigorous exercise per week. A regression of BMI on weekly exercise hours gives an intercept of 28.4 kg/m². What does the intercept mean here?

- A. The average BMI among adults reporting no vigorous exercise
- B. Nothing: the intercept is never interpretable
- C. The number of hours at which BMI reaches zero
- D. The amount BMI falls for each additional hour of exercise

Q9

A model of BMI on sleep hours reports $R^2 = 0.02$. Which conclusion is most defensible?

- A. Sleep accounts for about 2% of the variation in BMI, which is unremarkable for an individual-level health outcome
- B. The model is broken and should not be reported
- C. The model's predictions are 2% accurate
- D. 2% of people in the sample fall exactly on the line

Q10

A lab discovers its monitor reads 5 mmHg too low, so 5 mmHg is added to **every** systolic BP reading y_i . The ages x_i are untouched and the model is refitted. Compared with the original fit, what happens to $\hat{\beta}_0$ and $\hat{\beta}_1$?

- A. Both increase by 5
- B. $\hat{\beta}_1$ is unchanged and $\hat{\beta}_0$ increases by 5
- C. $\hat{\beta}_1$ increases by 5 and $\hat{\beta}_0$ is unchanged
- D. Neither changes, because a constant shift cancels out

Answer key

Do all ten first.

Q1. B. (*three jobs a model can do*) The model is being used to anticipate a future value from a new x , which is prediction. Description would be summarizing the relationship in past data; explanation would be claiming that changing the temperature changes admissions.

Q2. D. (*regression to the mean*) Selection on an extreme outcome guarantees that luck contributed to it, so the next measurement drifts back toward the athlete's true average. This is the same mechanism as the clinic that re-measures its highest-BP patients, and it is why single-arm before-after studies are so misleading.

Q3. C. (*confounding*) A confounder drives both variables. **B** describes an outlier and **D** describes the error term or residual: correct vocabulary, wrong object.

Q4. B. (*association and causation*) Adjustment moves you from level 1 to level 2 of the hierarchy, and no further. The honest sentence names the variables you adjusted for and admits that unmeasured confounding remains possible. **C** is the most common overreach in write-ups.

Q5. D. (*the least squares criterion*) A line that misses badly in both directions can still have residuals summing to zero. In fact every least squares line with an intercept has $\sum e_i = 0$, so this criterion cannot distinguish good lines from bad ones. **B** describes an argument *against* squaring, not against summing.

Q6. C. (*computing the slope*) $-0.30 \times 6.0/1.5 = -1.20$. **D** is $r \cdot s_y/s_x$ with the standard deviations swapped in your head, and **B** is $r \cdot s_y$ without dividing. Keep the units attached and the arithmetic checks itself: kg/m^2 divided by hours gives kg/m^2 per hour.

Q7. A. (*interpreting the slope*) Number, units, direction, “on average”, and a comparison *between* people. **B** is causal, which this design cannot support. **D** confuses a slope in the original units with a percentage change.

Q8. A. (*the intercept*) The intercept is always the model’s average y when $x = 0$. What makes it *interpretable* here is that $x = 0$ is a real, observed group: adults who do no vigorous exercise. Contrast this with a model of BP on age, where $x = 0$ means newborns and lies far outside the data, so the intercept is only a mathematical anchor. **C** describes where the line crosses the x -axis, which is a different quantity, and **D** is the slope.

Q9. A. (*r-squared*) Small R^2 values are normal for individual-level health outcomes, where most variation comes from biology and behavior you did not measure. A small R^2 does not make an association unimportant, and it does not make the model wrong.

Q10. B. (*changing the measurements*) Shifting every y by a constant slides the whole cloud of points up without tilting it. Formally, $\hat{\beta}_1$ depends on y only through $y_i - \bar{y}$, and adding 5 to every y_i also adds 5 to $\bar{y}$, so the deviations and the slope are untouched. The intercept then moves with the mean: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ gains exactly 5. **D** is tempting if you remember that “constants cancel” in the slope, but the intercept is precisely where the constant does not cancel.

One more thing worth knowing

This did not fit as a question, but it is the mistake that costs the most marks on the interpretation questions, so it is worth thirty seconds now.

Compare these two sentences about a slope of -0.32 for sleep hours:

Adults who sleep one hour more have, on average, $0.32 \text{ kg}/\text{m}^2$ lower BMI.

Sleeping one more hour lowers BMI by $0.32 \text{ kg}/\text{m}^2$.

The first compares *different people* at one moment, which is what cross-sectional data contains. The second describes what happens *inside one person* over time, which the data never observed. The numbers are identical and the claims are not.

When you write an interpretation, check that it contains four things: the number, the units, the direction, and a phrase such as “on average” or “is associated with”. If the sentence would still make sense as an instruction to a patient, it is probably too strong.

Systematic changes to x or y

Question 10 is one case of a pattern worth knowing in general. Everything below follows from the two formulas you already have,

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Change to the data	New slope	New intercept
Add a to every x_i	$\hat{\beta}_1$	$\hat{\beta}_0 - a\hat{\beta}_1$
Add b to every y_i	$\hat{\beta}_1$	$\hat{\beta}_0 + b$
Multiply every x_i by c	$\hat{\beta}_1/c$	$\hat{\beta}_0$
Multiply every y_i by d	$d\hat{\beta}_1$	$d\hat{\beta}_0$

Two things to take from the table. **Shifting** either variable never changes the slope, because the slope is built from deviations around the mean and a shift moves the mean by the same amount. **Rescaling** changes the slope by exactly the factor needed to keep the fitted line in the same place: measuring age in decades rather than years divides x by 10, so the slope is multiplied by 10 and describes a decade instead of a year.

In every row the fitted line is the same line, drawn on relabelled axes, so the residuals are unchanged up to the same rescaling and R^2 does not move at all. Only the description changes, which is why units belong in every interpretation you write.